Lab: Exploring Cognition in Microstructural Evolution

Objectives

> **Learning Goal:** Apply the Active Inference concept of cognition to a concrete metallurgical scenario using image analysis lab techniques.

This is a **image analysis lab exercise** designed for materials scientists and engineers.

Part 1: System Definition (10 min)

Choose a metallurgical system to analyze. Suggested options:

- Fe-0.4wt%C steel undergoing austenitization and cooling
- Al-4wt%Cu alloy during precipitation aging
- Ti-6Al-4V produced by laser powder bed fusion
- Pure copper undergoing recrystallization after cold work

Write your response here

Part 2: Cognition Analysis (25 min)

2A: Identify the Active Inference Components

Map your chosen system onto the Active Inference framework in the context of cognition:

Component	Your System
System boundary (Markov blanket)	
Internal states	
External states	
Sensory states	
Active states	

Write your response here

2B: Cognition Dynamics

Describe how cognition operates in your system:

- What is the driving force (free energy gradient)?
- What is the timescale?
- What characterization technique would you use to observe it?

Write your response here

Part 3: Quantitative Exercise (20 min)

Perform a simple calculation or simulation related to cognition in your system:

- If computational: use Python with PyCalphad, NumPy, or similar

- If analytical: use thermodynamic relations (Gibbs energy, diffusion equation, nucleation barrier)

Write your response here

Part 4: Reflection (10 min)

1. How did the Active Inference framework change your understanding of cognition in this system?
2. What prediction error exists between the equilibrium model and the real behavior of your system?
3. How could you reduce this prediction error through better characterization or modeling?

Write your response here

Summary

Analysis Component	**Key Finding**
System analyzed	
Cognition mechanism identified	
Driving force	
Main prediction error	
Recommended next characterization	

Write your response here